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ABSTRACT

We will describe a method for allowing the privacy preserving trust minimized exchange of cryptographic assets across blockchains.

Currently, the two most popular methods for exchanging crypto-assets are centralized off-chain exchanges and Over-The-Counter (OTC) Brokers. Of the two, exchanges are the most well-known and popular. However, centralized exchanges are fundamentally unsafe because they take custody of their users' assets. This also presents an existential risk to the crypto-asset community at large. by centralizing asset custody more than necessary.

The creation of noncustodial exchanges solves both issues by keeping assets in direct control of their holders while allowing exchanges to continue providing value to customers and the community at large.

Ex_net is a federated and bonded Proof of Stake(PoS) network which creates an electronic venue for the noncustodial federated cross chain exchange of cryptographic assets. It provides all the back-office features one would expect from a modern cash foreign exchange (cashFX) platform, with an emphasis on efficient order matching, price discovery, and the preservation of privacy.

In addition to facilitating P2P and noncustodial exchange, ex_net also allows for the off-chain creation and distribution of liquidity proofs, which can be used to prove to third parties that assets are locked on their respective

blockchains and available for trading. The efficient creation and distribution of these proofs provides new levels of transparency and privacy to the users of ex_net. The tools and services provided by ex_net mature the crypto-asset market while providing long desired features unavailable in any existing trading platform.

This document will discuss the network entity roles, features, and underlying blockchain architecture of ex_net.

INTRODUCTION

The Bitcoin Protocol was not designed with a method for linking the network intrinsic value of bitcoins to network extrinsic value. Because of this, the early days of bitcoin required in person transfers of bitcoin for fiat (or some other physical asset like pizza). Given the electronic nature of Bitcoin, it wasn't long before internet based centralized exchanges emerged to facilitate the exchange of fiat for bitcoins.

The addition of these out-of-protocol custodial intermediaries introduced new risks to both the users of the services and the blockchains which those services support.

If we instead make a federated, permissioned blockchain with validation rules, tuned exclusively for matching orders and exchanging assets, we can have the transparency, auditability, and fault tolerance of a PoS blockchain while very closely approximating the speed and throughput of traditional databases.

Furthermore, if we apply cryptography correctly, we can also allow traders using such an exchange to decide which privacy tradeoffs they want to take and when. This cryptography will also allow participants to provide *transitive* cryptographic proof of liquidity. E.g., Counterparties on ex_net can provide proof of liquidity across any number of parent chains. As an added bonus, operators of on-chain matching engines can prove to their traders that matching occurred as prescribed.

Cryptographically blinding fields in orders provides several benefits to traders. Partially blinded orders are an extreme deterrent to front running and allow traders finely tuned control over who can fill and route their orders, even when orders are being matched via ex_net's on-chain matching engines.

Ex_net functions as a venue for all coins and protocols supported by the ex_net validators. The term decentralized exchange is popular today. As a verb this is fine, but as a noun it is inaccurate. the system that we are proposing is a federated exchange venue, venue meaning all the additional services needed to facilitate /execute trade. This includes asset tracking, simple identity management, price and counterparty discovery.

Fairness and transparency on ex_net are improved by placing matching engine state changes in blocks. Doing so acts as a "speed bump" and ensures that all network participants have a fair opportunity to see all state transitions within the venue. These state changes include the change of exchange rates between all pairs traded on the network. Blocks on ex_net contain more than just matched orders, they also include the state of the parent chains ex_net uses to settle filled orders.

Ex_net is operated by a federated and bonded pool of validators. These

validators are responsible for maintaining ex_net as well as ensuring transparency and auditability. Ex_net validators provide an accurate view into the state of all parent chains even when those chains are experiencing re-orgs. These require that validators on ex_net be full nodes on parent chains.

Validators also provide parent chain settlement of completely or partially filled orders via payment channels. To provide the highest level of transparency to traders using ex_net, the validators will provide third-party auditable snapshots of parent blockchains every block. These snapshots will then be used by traders to generate liquidity proofs.

Liquidity proofs are used by traders to provide ephemeral liquidity, a novel solution to the problem of ghost liquidity on traditional stock exchanges. The trader offering liquidity specifies a minimum time and maximum time that they will make the liquidity available. These time ranges are enforced by ex_net validators.

The limitations of known physics and the physical configuration of the network, legal agreements, and on-chain governance (including parent chain hosted bonds) all work together to ensure that traders can trust the validators on ex_net are following the strict order matching rules they claim to be. The transparent nature of the chain ensures that traders will be able to determine within one block if validators have allowed for front-running or other sorts of malicious behavior.

A trader's connection to the network is vitally important in ensuring they have an accurate view of the venue, but because of block latency, it is much easier for traders to know they are in sync with their trading peers.

As with any blockchain, governance is of utmost importance. In keeping with the open source culture from which blockchains emerged, all network governance will happen on-chain. This added transparency allows users to minimize the trust required to confidently transact with counterparties over the network.

Although it would be possible to operate ex_net as a single entity, operating it as a federation provides numerous benefits to all entities using the network. Often when designing blockchain-based systems there are convoluted discussions around economic incentives. For this document, we will only point out that traders make and take orders because it benefits them to do so, and ex_net validators take a percentage of the assets exchanged in every filled order. Ex_net validators may also choose to create and fund a legal entity tasked with maintaining the incentive alignment between individual and collective validators, and the traders that use the network.

Since locality is always a factor in the cost of exchange, we can expect multiple ex_net exchanges to exist in the world.

In the rest of this document, we will describe how the ex_net protocol can facilitate a “pure” crypto-asset exchange venue with tunable privacy and intermediation.

There are a few features that have been considered and deemed technically possible, but too complex to be sufficiently covered in this single document. Those features are:

- The Issuance of network intrinsic Promise(Debt) back (PBTs)
- Zero Knowledge Proof (ZKP) based features such as Transaction Field Blinding and Identity proofs.

Debt Issuance will be completely omitted. Instead of ZKPs we will assume more simplistic secret sharing protocols.

Network Features

DESCRIPTION

Noncustodial cross-chain asset exchange is currently unavailable on any popular exchange and in extremely high demand. Ex_net addresses this issue by creating a federated and bonded Proof of Stake Network and Protocol. Payment Channel and HTLC technology on existing chains allow counterparties to engage in non-custodial/trust-minimized cross-chain cash FX-like transactions. The protocols that exist today-Lightning, Interledger, and Raiden-, are only partially compatible and insufficient for creating a fully P2P venue. The venue created by ex_net is noncustodial and chain agnostic/cross-chain.

Ex_net is a blockchain linking existing blockchains. If we wish to maintain the trust-minimizing properties of blockchains while exchanging crypto-assets across two or more chains, the optimal method is the creation of a blockchain which operates “on top” of the parent chains.

The two parent chains assumed throughout this document are Bitcoin and Ethereum. Furthermore, this document assumes the Lightning protocol will support Bitcoin-style and Ethereum-style networks in the near future. Ex_net will be using Lightning payment channels and Lightning hashlocks across all parent chains supported by Lightning.

Trading: Unlike a simple payments system, a network for exchange requires that traders are able to find and address each other; this requires simple identities on the network. Ex_net will provide traders with identity services required to facilitate trading in whichever jurisdictions it operates. Ex_net also provides an open order book, price curve feeds and Lightning-based cross chain P2P/OTC exchange. Orders are backed by assets locked in payment channels on parent chains. In order to provide a quality experience to its users, ex_net will also

provide the externally verifiable cryptographic evidence required to prove the existence of identities and locked assets across all parent chains. Ex_net validators provide a per-block exchange rate, called a price curve, for each pair offered on the exchange. This price curve is calculated from looking at the history of executed trades on the network for the purposes of helping traders set prices optimally. When traders wish to create a limit order, they set the price curve in the order, otherwise they will create market orders by using the price curve provided by the validators. When ex_net validators deem it necessary, they will provide additional support for ERC20s and bitcoin color coins. If ex_net decides to support interledger, it will build its own bridge.

Connecting to Validators: Validators should connect to each other in an internet exchange. Node ownership and operation requires technical sophistication that most traders do not have, so traders will connect to the validators via a web interface provided by the validators. Liquidity providers may connect via either the web interface or they can run their own full nodes and connect to ex_net. Validators can decide amongst themselves what quality of service guarantees are provided to traders and liquidity providers.

Tokens and Vouchers: Users on the network can transact with tokens on parent networks, such as Bitcoin and Ethereum, via payment channels. Users also have the option of depositing crypto-assets with ex_net validators and receiving vouchers on ex_net. Ex_net does not have a native token. Unlike services such as BTCRelay, users can redeem vouchers on ex_net for any crypto-asset at any time.

Wallets: Traders and liquidity providers maintain their own wallets, which allow them to have custody of their cryptographic assets. By default, this wallet will only interact directly with ex_net. If users of the wallet so choose, the wallet can optionally connect with any Lightning, Bitcoin, or Ethereum node, allowing them to verify the accuracy of the information provided by ex_net. Users can also import private keys from other networks and link them together in a privacy preserving way, simplifying the process of trading across blockchains.

Liquidity Proofs: A trader using ex_net has the ability to generate from their wallet a liquidity proof, which is simply a collection of transactions, potentially across multiple blockchains, showing that a particular trading pair or portfolio of assets is guaranteed through a collection of cryptographic signatures and on-chain messages liquid for some period-of-time/number-of-blocks.

The proof does not require the signatures of validators, but instead provides a succinct path of traversal through caches which are generated, maintained and signed by validators.

ENTITY ROLES

An entity is anything that acts from within the network. An entity may be a real-world company, person, piece of software, or a combination thereof operating together. Any number of entities can own any number of accounts. These accounts take on various roles within ex_net. From the perspective of the protocol, there exist bonded validator accounts and accounts which do not validate.

As is the case with any blockchain, wallets are how entities interact with their accounts. Users on the network use ex_net wallet software to interact with caches generated by validators. These caches are used to create liquidity proofs and track the state of assets across various networks.

For the purposes of discussion, we will consider three roles within the network: traders, liquidity providers, and validators. Users discussed here are traders and liquidity providers.

TRADERS

These are entities wishing to exchange one cryptoasset for another. Typically, a trader's order can be filled completely within a block. In addition, traders generating Liquidity Proofs may blind or unblind various fields within the liquidity proof. The ex_net wallet will provide a view into the state of ex_net and parent chains, which should satisfy the safety requirements of most traders.

LIQUIDITY PROVIDERS

These are entities who are sophisticated enough to keep payment channels open for amounts larger than what can be filled in a single block. They are long term holders, but not validators. Liquidity providers may want a more detailed view into ex_net and will run full nodes in addition to wallets.

VALIDATORS

A federated, bonded pool of validators will retain significant control and governance over the network, along with its attendant responsibilities.

In order to enact the numerous parameters needed to operate this network with the intended level of safety, consensus is needed among the validators. Ex-Net requires validators to not only coordinate across the PoS network they are operating, but to attest to the state of other blockchains.

VALIDATOR RESPONSIBILITIES

Validators on ex_net are responsible for facilitating the trading of cryptographic assets between ex_net wallet holders. It is possible for ex_net validators to allow trading from outside of ex_net wallets, but this is outside the scope of this document. Ex_net will use Tendermint for its consensus protocol.

There are multiple features for which ex_net must provide a trust minimized

environment to its wallet holders. Notably, a trader's cryptographic assets are never in sole custody of ex_net or an individual ex_net validator. Instead, traders maintain custody of their assets on whichever parent chain they came from.

Bonding

Additionally, to minimize the trust traders must have in a single validator, validators are bonded with each other through bonding contracts on the parent chains, Ethereum and Bitcoin. These bonding contracts use payment channels. For each parent chain that validators use to open payment channels with traders, the validators must be bonded to each other on that blockchain before they open payment channels with traders. Collectively and individually, validators are responsible for maintaining the state of all their payment channels across all networks.

As validators are added and removed from the chain and their balances are changed due to further funding or slashing, the validators are responsible for making sure there is a sufficiently larger bonded balance to facilitate projected trade activity. Validators link various bonds across various chains with their identities on ex_net by signing parent chain public keys with ex_net private keys and publishing the result in blocks on ex_net.

Block Creation

When blocks are proposed on ex_net, the proposer inserts a set of state changes for every parent chain into the block. Each validator is responsible for validating these state change sets. For each parent chain, quorum needs to be reached on the state change set proposed. If quorum is not reached for a state change set, trading across that parent chain halts. If the proposed block contains no valid state change set, then the block is rejected by the other validators and trading halts while a new leader is selected. Blocks on ex_net are restricted in size based on the amount of value exchanged in the matched and filled orders contained in the block. There can never be more assets matched and/or filled in a block than the total assets at risk if the validators equivocate when signing that block.

Connecting to Traders and other Networks

Each ex_net validator is responsible for maintaining their own connection to all parent chains. They are also required to track the state of open payment channels on all parent chains. If they so choose, all validators are also required to track the state of tokens on parent chains, such as ERC20s on Ethereum or Omni tokens on Bitcoin.

Depending on the relationship between the trader and ex_net validators, the trader may wish to send their order to a single validator with which they have either a bond on a parent chain or a legally binding agreement. The trader may also choose to broadcast the order to a quorum of ex_net validators who are responsible for providing a receipt.

Validators are responsible for maintaining the open orderbook and

matching orders in other account orderbooks. They will use a strict order matching algorithm which will rely on entropy inside transactions to determine the sequence of order execution.

When orders in the mempool match with each other or with an existing open order, a fill message is generated and placed into the proposed block. A match, fill and settlement occur in the same block if the fill is composed entirely of vouchers. Otherwise, after the block containing a fill has been signed by a quorum of validators, the signing validators will authorize settlement. Settlement occurs when assets are exchanged between counterparties via payment channel or when voucher balances on ex_net are updated.

P2P settlement may happen asynchronously from block creation such that it references a fill that has not yet made it into a block.

Additional conditions which halt trading

A network split on ex_net would cause trading to halt. A prolonged fork will stall trading on the forked parent chain. If there is an extreme devaluation of the parent chain bonds made by validators, throughput on the network will naturally slow. The settlement of an exchange between two traders requires that enough validators are available to unlock the requisite assets. Ideally, in the case of vouchers and P2P payment channels, 2/3rds of ex_net validators are needed to facilitate the exchange. In the case of parent chain payment channels, the majority necessary varies, depending on the state of the bonded validators, there are situations where ex_net has failed but validators can still safely close payment channels on the parent chain.

ORDERS

WHAT IS AN ORDERBOOK?

Orderbooks are where orders are sent by traders to be matched. First, a user's ex_net wallet generates accounts according to security parameters set by the user. Validators then generate an orderbook per account. The ex_net wallet collects all the orderbooks for the accounts it created and compiles them into a view of all orders only visible to the user. Orderbooks are composable; an orderbook can be a trader within any number of orderbooks. Orderbooks advertise fees, which are taken by validators based on successful order filling.

The orderbook contains matching rules, routing rules, and fee schedules. These are tunable such that different accounts can have different fees for different trading pairs. Traders can also block trading to individual accounts such that their engine will not match the orders of those accounts.

The validator's manage a single orderbook amongst themselves called the Venue's orderbook.

WHAT IS AN ORDER?

Ex_net traders exchange assets with each other via orders. There are two types of orders available on ex_net, backed orders and unbacked orders. The term “orders” in this document refers exclusively to backed orders; unbacked orders will be supported by the protocol but are out of this document’s scope.

An order is backed when the assets to be exchanged are escrowed in the order itself. There are three options available for backing an order: ex_net vouchers, open orders, and assets held in payment channels on parent chains. For any given payment channel, there can be only one *open* order. For any order, there can be only one payment channel.

There are three types of orders that can be placed using the ex_net wallet:

- Unblinded orders, which are sent to the pool of bonded validators to be matched
- Blinded orders, in which values can be encrypted to one or more parties by the trader and sent to validators. Anyone can see that the order was made, but only the sender and receiver can see the order values. The receiver’s wallet needs to be online to decrypt a blinded order, this may or may not require human intervention.
- P2P orders, which occur off-chain and utilize the Lightning network’s privacy preserving payment channels.

HOW IS AN ORDER CREATED?

Once a trader has fully configured their wallet with ex_net, they can create orders on ex_net. An order is valid if it is backed by assets locked in a payment channel between ex_net and the trader on a parent chain. An order can also be backed by vouchers on ex_net. Using the payment channel or vouchers as proof of backing, a trader can offer to trade the backing asset for any number of other assets that are registered for exchange on ex_net. Within the order, traders are able to set, the rate of exchange (price curve) between their backing asset and any assets they will accept in exchange.

Orders have a minimum and maximum number of blocks that must be available within a orderbook. Within this block range, traders are required to set an expiration time for their order. This is crucial because orders must be removed from the orderbook and the orderbook charges a fee the longer a order sits on the book. The finite duration of orders restricts the manner in which they can be routed. That fixed duration is in part derived from the expiration properties of the payment channel backing the order. Traders have two ways to restrict where their orders can be decrypted: symmetric key cryptography and where they send the order.

Traders can restrict with whom an order can be filled by specifying a list of valid fillers in the order. Traders can specify a list of which fillers can route the order. A well formatted order contains fields encrypted such that the filling and routing specified in the order is possible. The wallet of the order's receiver is responsible for determining the order validity and steps to be taken if the order is invalid. Fees for the order are calculated when the order is constructed based on values provided by the validators. If the fee for filling the order is more than the value of the order, the order is rejected by the engine running on behalf of the receiver.

HOW ARE ORDERS MATCHED AND FILLED?

Once a trader has created an order, they have the option of sending it directly to an account the order specifies as a filler. If the order specifies no accounts, then it's an open order matched by the validators. As mentioned above, there is a default matching rule that validators apply to sent orders. Accounts can also provide on-chain custom rules such that they can optionally match orders off-chain. Multi-sig accounts can be created that have their own matching rules.

The default matching rules used by the validators to operate the matching engine allow for the partial matching of orders.

HOW ARE ORDERS ROUTED?

When creating an order, a trader decides which orderbooks they would like to match it, and sends the orders directly to those orderbooks. Blinded orders sent to the blockchain can be used as privacy preserving proof of backing for off-chain matching and filling. Routing a blinded order requires that the receiver decrypt it off-chain.

If the filler can route the order, they create a derived order which has the order as proof of backing. The derived order must expire before the primary order so that the proof of backing remains valid. As is the case with the primary order, the recipient of the derived order is responsible for determining if the derived order is valid before attempting to fill it. This derived order verification can become complex if the fields in the order are blinded.

HOW IS AN ORDER FILLED/SETTLED?

Unlike any other exchange, when the order is filled on ex_net, the only way the fill can fail to be converted into settlement is if a quorum of ex_net validators are unable to authorize the settlement across the parent chains involved in the fill within the timelock on the backing assets.

In an ex_net fill, ex_net validators record a matching event for a set of orders in an ex_net block. Settlement is where payment for a filled order is made via P2P or parent chain payment channels. This document will discuss two types of orders: orders backed by other orders and orders backed by assets, derived orders and primary orders, respectively.

An open primary order can be open due to an unsettled derived order. If this is the case, the derived order must be settled for the primary order to be

settled

WALLET FEATURES

If an entity wishes to interact with ex_net, they must use an open source wallet provided by the ex_net validators. This wallet supports all the networks/blockchains ex_net supports, for example, ex_net, Lightning, Ethereum, and Bitcoin. An ex_net trader creating an order for BTCs would use their ex_net wallet to place BTCs in a payment channel with ex_net validators. If the asset the trader requests is on Ethereum. Ex_net validators can fill that order through a Lightning-on-Ethereum payment channel. Ex_net wallets will also use the Lightning network to facilitate P2P orders. Off-chain matching may or may not be supported by ex_net wallets and can occur via a Lightning network HTLC-compatible hub spoke model.

Address signing/Identity proofs

There are three primary ways that assets from parent chains can be managed within the ex_net wallet. The first way is to complete a simple send/transfer on the parent network. A trader has assets in an address on a parent chain that they wish to trade on ex_net. The trader creates an address for that parent chain in their ex_net wallet, and then sends assets from their parent chain wallet to their ex_net wallet.

However, a trader may instead import the private key from the parent chain into the ex_net wallet.

There may be times where a trader wishes to attest to some balance in a wallet on the parent chain. In this case, they need to sign a message with their parent chain private key, which includes their ex_net public key, and publish that message to ex_net.

In future versions of ex_net, users will be able to create zero knowledge proofs to securely identify themselves to counterparties.

Asset Tracking

Ex_net validators provide cryptographically signed views into both ex_net and the parent chains. Via their ex_net wallets, traders can see their assets move in/out of ex_net and to/from parent chains in real time. The wallet also manages the trader's P2P payment channels.

Liquidity Proofs

Because of the privacy preserving nature of the ex_net wallets and protocol, it is impossible for the ex_net validators to attest directly to a trader's assets.

Liquidity proofs are a collection of cryptographically signed statements showing to a counterparty which assets a trader has available for trade on ex_net and under which constraints. Traders can generate liquidity proofs via

their ex_net wallets. Any information that the trader has published to ex_net can be included in a liquidity proof, including attestations to assets outside the ex_net wallet.

Viewing Orderbooks

Validators generate price feeds from the matching engines they host, such as those for the open orderbook or trader orderbooks. Prices from P2P orders, the Venue's orderbook and account orderbooks are available to traders via the ex_net wallet in real-time.

Because of the privacy preserving nature of ex_net orders, each trader may have a unique price curve available to them for every trading pair at every block.

Anyone can see the Venue's orderbook price of an asset on ex_net whether or not they have a wallet on it.

Modifying Matching Engines

Traders are able to modify their custom matching engine rules via the wallet interface. From this interface they can filter other users, asset types, or turn off the on-chain matching engine completely using fills sent to their orderbook from off-chain to match orders.

PHYSICAL CONSIDERATIONS

Although we may sometimes forget, computation and in turn electronic exchange is constrained by physical laws. It will always be the case that traders who take a more direct path to the matching engine will have an advantage over those who do not. When considering the problem of decentralized exchange as a verb, we must accept these physical constraints on decentralized exchange as a noun.

A truly decentralized exchange will by definition enable operators to front-run. Thus, federated exchanges allow traders to hold venue operators responsible for providing a fair platform in the most trust minimizing way possible.

This puts the onus of creating a robust physical network onto the bonded validators.

Geographic Locality

There are a number of factors necessary to facilitate the operation of a federated exchange.

Federated members must be confident in their ability to punish each other, at a minimum through bonds placed on parent chains. They may also want the added protection of being able to sue each other.

Thus, this ability to sue each other applies some constraints on the geography of the exchange. Additionally, if the physical distance between the validators is too great, the network will not be able to achieve the extremely low block times required to support the maximum number of parent chains.

Internet locality

In addition to the validators needing high quality interconnectivity, individual and collective validators need high quality connections to all parent chains serviced via ex_net. The best way to achieve this high quality internet connectivity is to co-locate ex_net in an internet exchange. Individual validators, or small sets of validators, may choose to create a dark fiber connections to traders or liquidity providers with whom they have special relationships.

Hardware

Ex_net validators will improve their collective profits by processing as many transactions as possible. Because each parent chain has a limited amount of throughput, payment channels are vital in allowing ex_net to process as many transactions as possible. Additionally, each additional parent chain is another opportunity to increase transaction volume. Scanning parent chains for decentralized exchange activity and opportunistically filling those orders is another opportunity for ex_net validators to maximize transaction volume.

In order to engage in all these activities with a high level of synchronicity and exceptionally low block times, validators must operate a resilient stack of high performance hardware at the venue.

How validators decide exactly which hardware stacks should be supported is a question that must be resolved amongst themselves via the on-chain governance mechanisms.

GOVERNANCE CONSIDERATIONS

As previously mentioned, federation validators may wish to employ both crypto economic and legal methods for ensuring the federation's integrity. It's also important that federation members have some level of insight into each other's business operations. Large trading customers who have not yet transitioned into becoming validators themselves may wish to maintain relationships regarding order flow with multiple validators. Decisions regarding software, hardware, internet connectivity, bonding changes, and adding or removing validators all must occur on-chain via an on-chain governance system.

This allows for traders using the network to be aware of any changes that may impact their ability to trade in the future.

LEGAL STRUCTURE

The validators can organize themselves under a variety of legal structures. These legal structures interrelate to geographic, on-chain governance, and internet connectivity considerations. One point of contention may be what to do with the assets that are slashed when validators equivocate. One possible solution is

to place these assets in a trust to further the open source development of the parent chains, Lightning network, and open source wallet software. Alternatively, these funds may be donated to a charity, but proving that the donation recipients do not represent an attack against the network implies an onerous burden on the validators.

HOW VALIDATORS SELL THEIR SERVICES

Validators sell their services via being existing exchanges.

CONCLUSION

This document provides a clear high level explanation of the ex_net protocol. it also touches on some of the considerations required to take such a system to production. Future documents will provide more depth into the cryptographic protocols and software implementation details required to have such a network perform optimally.